

	NFC	RFID	UWB	Bluetooth
Set-up time	<0.1ms	<0.1ms	~0.5s	~6 sec
Range	Up to 10cm	Up to 3m	Up to 5m	Up to 30m
Usability	Human centric Easy, intuitive, fast	Item centric Easy	Data centric Easy	Data centric Medium
Selectivity	High, given, security	Partly given	Line of sight	Who are you?
Use cases	Pay, get access, share, initiate service, easy set up	Item tracking	Control & exchange data	Network for data exchange, headset
Consumer experience	Touch, wave, simply connect	Get information	Easy	Configuration needed

Fig. 1: Courtesy – Scolis Technologies

synchronisation for data transfer. It, however, does not exhibit inherent security. But NFC, which operates in the radio frequency of 13.56 MHz within a distance of 10 cm, provides a high degree of security due to its short range. The communication between two NFC enabled devices takes place through magnetic field induction in an air-core transformer that is formed between the antennae of both the devices.

Application modes

Peer-to-peer model. The peer-to-peer communication model allows you to transfer data between two devices using short-range wireless technologies, for example—Infrared (IR) and Bluetooth.

For infrared communication, IR ports of two communicating devices like mobile phones or laptops need to be in the line-of-sight. In Bluetooth data transfers, you don't need line-of-sight, but you have to undergo the tedious process of configuration setting, searching of another Bluetooth device and then selecting it.

NFC, however, combines the convenience of both IR and Bluetooth. It does not require the line-of-sight or setting configuration, searching and then selecting another device, yet provides a secure channel for data communication.

When you bring two NFC-enabled devices near each other, a secured p2p communication is established that can be used for fast, easy and convenient

data transfers.

Tag reader or card emulator.

Consider a situation where you come across a 'smart' poster of a live concert happening in your city. Being a smart poster it incorporates an NFC tag that has all the details of the event. When you bring your NFC-enabled cellphone close to it, it reads and captures all the information.

Your NFC-enabled cellphone can also detect a phone number and ask you to initiate a call to the number, or open the Internet link directly on your mobile Internet browser.

The card reader mode of NFC cellphone is useful for content distribution, information access, etc, and can be linked to other applications like initiating a call or opening the Internet link through mobileweb.

In card emulation mode, the NFC device acts as a tag itself with some

stored information. For example, you can store all your credit card details on your NFC cellphone and make payments by scanning it on NFC readers.

Targeting mobile phones

NFC is primarily aimed at the mobile phones. It is best suited for them, as they are capable of downloading information, songs, ticket information, top-up a travel card, banking applications, etc. And, the most important fact is that NFC readily fits in mobile handsets.

"In more than 30 NFC field trials in various countries, we have observed that consumers show a strong desire to use their cellphones for mobile banking, money transfer, payments, shopping savings/ discounts, and shopping assistance," says Mohammad Khan, president, Vivotech.

"More than 75 per cent of consumers responded positively after participating in mobile payment trials. They prefer to use their NFC phones to pay at participating merchant locations. This is because of convenience factor and easy accessibility of payment instrument through mobile phones. They also find mobile phone to be a dynamic device with a user-friendly interface as compared to a passive card," he adds.

Application areas

Initiating a connection. Data transfer can take place between an active pair of NFC devices. You can display pictures from your NFC digital camera by bringing it close to your digital

TechFacts

NFC technology specifications

- A short-range wireless communication – 10 cm
- Standards based (RFID 13.56MHz, ISO 14443A/B, 18092, ETSI, ECMA)
- Consumes low power
- Speed – 106, 212, 424 kbps

NFC Forum specifications

- NFC data exchange format (NDEF) defines common data format for NFC compliant devices
- Four specific record type definitions (RTDs) for building standard record type
- Four tags type (type 1/2/3/4), which enables core interoperability
- Connection handover defines structure and sequence of interactions between NFC devices